

Indexical Shift and its interactions with Perspective Sensitive Items (PSIs): Indexical PSIs in Turkish

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Abstract The link between indexical shift and perspective shift is not well-understood. In indexical shift, indexicals nominals (e.g. *I, you, here, tomorrow*) are interpreted relative to a reported context, rather than the actual context of utterance. In perspective shift, perspective sensitive items (PSIs) (e.g. *tasty, local, foreigner*), are interpreted from the viewpoint of an individual other than the speaker. This study examines how PSIs and indexical shift interact in Turkish attitude reports, and shows that PSIs do not exhibit a uniform behavior. In light of novel data, this study shows that some PSIs, termed *Indexical PSIs* (e.g. motion verb *gel* ‘to come’, evidential markers), show shift together patterns with indexical nominals, being targeted by indexical shift. In other words, Indexical PSIs must shift when they appear in an indexical shifting context, and cannot shift without indexical shift. Other types of PSIs, on the other hand, do not interact with the occurrence of indexical shift. The existence of Indexical PSIs supports context over-writing approaches to indexical shift over alternative accounts that try to reduce indexical shift into simple feature deletion/impoverishment.

Keywords: indexical shift; perspective shift; perspective sensitive items; Turkish; attitude reports

1 Introduction

Indexical elements such as English *I, you, here* and *yesterday* get their interpretations relative to the context in which they are uttered (Kaplan 1977). For example, if (1) is uttered by *John* in *Los Angeles*, it means that John was born in Los Angeles, but if it is uttered by *Mary* in *Boston* it means that Mary was born in Boston.

(1) I was born **here**.

Indexicals refer to the referents of the speech-act coordinates like *author, addressee, location* and *time* (e.g. Kaplan 1977; Schlenker 2003; Anand & Nevins 2004; Deal 2020). For example, *I* is used to refer to the *author* (speaker) of the utterance, while *here* is used to refer to the *location* where the utterance was made, and thus their reference is determined based on the actual context of utterance. However, there is a relatively large literature reporting that some languages, including Amharic (Schlenker 1999), Zazaki (Anand & Nevins 2004), Slave (Anand & Nevins 2004), Uyghur (Shklovsky & Sudo 2014), Nez Perce (Deal 2020), and Turkish (Şener & Şener 2011), allow indexical shift, a situation where an indexical element gets its referent from the reported context, rather than the actual context of utterance. For example, the Uyghur 1st person indexical *men* ‘I’ and the 2nd person indexical *sen* ‘you’ in (2) refer to the speaker and the addressee of the reported context, *Ahmet* and *Aygül*, rather than the actual speaker and addressee. Though indexical shift is similar to direct quotation in that it shifts how indexicals are interpreted, it is important to note that indexical shift and direct quotation are different phenomena (see Section 2 for details).

- (2) Ahmet Aygül-ge [*men seni yaxshi kör-ymen*] di-di.
 Ahmet Aygül-DAT 1SG 2SG.ACC well see-IMPF.1SG say-PST
 ‘Ahmet_i told Aygül_j that he_i likes her_j.’
 (Uyghur, from Shklovsky & Sudo, 2014, pp.395)

Indexical shift overwrites the values of the *author* and *addressee* coordinates of the context, relative to which the sentence is interpreted. More precisely, the *author* and *addressee* coordinates of the embedded clause are replaced with the speaker and the addressee of the reported context (e.g. Anand & Nevins 2004; Sudo 2012; Shklovsky & Sudo 2014; Deal 2020), and thus in (2) the 1st person indexical *men* picks the attitude holder *Ahmet* as its referent while the 2nd person indexical *sen* refers to his addressee *Aygül*.

A similar kind of shifty behavior is observed with *perspective sensitive items* (PSIs), whose interpretation can be context dependent (e.g. epithets like *idiot*, predicates of personal taste like *tasty*, relative socio-cultural expressions like *foreigner*). PSIs are typically interpreted from the point of view of a *perspective holder*. For example, to be judged *true* or *false*, the sentence in (3) needs to be interpreted from the perspective of a person (e.g. Speas & Tenny 2003; Lasersohn 2005; Oshima 2006; Stephenson 2007; Bylinina et al. 2014; Anand & Korotkova 2021).

- (3) John is a *foreigner*.

In other words, the truth of (3) depends on where the perspective holder, the person under whose perspective *foreigner* is evaluated, is from. As Bylinina et al. (2014) put it, the sentence in (3) roughly means ‘John is a foreigner to the perspective holder’s country’. It is generally assumed that the perspective holder of a clause is represented with the *judge* coordinate of the context (Lasersohn 2005; Stephenson 2007; and subsequent work). In simple sentences like (3) the perspective holder is the speaker of the sentence by default (e.g. Lasersohn 2005; Bylinina et al. 2014; Anderson 2021; Kaiser 2022a). However, in some syntactic contexts like attitude reports, *judge* can shift to a different person than the speaker, the phenomenon known as *perspective shift*. When perspective shift occurs, the PSIs are interpreted under the perspective of a new perspective holder, rather than the perspective of the speaker. (4) illustrates that *perspective shift* can optionally occur in English attitude contexts embedded under the verb *to think*.

- (4) Mary thinks that John is a *foreigner*.
 a. non-shifted → ‘John is from a country other than where speaker is from’
 b. shifted → ‘John is from a country other than where Mary is from’
 (adapted from Bylinina et al., 2014)

Without perspective shift, the relative socio-cultural expression *foreigner* is interpreted under the perspective of the speaker (4a). When perspective shift takes place, *foreigner* is interpreted under the perspective of the attitude holder *Mary* (4b).

At least at first glance, *indexical shift* and *perspective shift* appear to share some common properties. For instance, they both replace one of the context parameters (*author* or *judge*) with the attitude holder, and they can take place in same syntactic environments (e.g. attitude reports). Despite these similarities, the relatively large literatures on these phenomena seem mostly disconnected, and the relation between *indexical shift* and *perspective shift* is not yet fully understood. Considering that *author* and *judge* coincide to refer to the speaker of the sentence in non-shifted (indexical or perspective) clauses, it might be the case that the *judge* coordinate

is sensitive to the value of the *author* coordinate, and it refers to the same entity with *author* by default. In that case, indexical shift might indirectly change the interpretations of PSIs by changing the referent of the *author* coordinate (i.e. indexical shift might force perspective shift). Alternatively, if *author* and *judge* coordinates are totally independent from each other, indexical shift and perspective shift might independently change the referents of the *author* and *judge* coordinates without affecting the other. It is also possible that there might be limitations as to how many ‘shifting’ operations can take place, and the occurrence of indexical shift might block the occurrence of perspective shift (or vice versa).

Recent work investigates these possible interactions between indexical shift and perspective shift, and suggests that these two phenomena are in a subset-superset relation, where availability of indexical shift entails availability of perspective shift (but not vice-versa) (Bylinina et al. 2014; Sundaresan 2021). Sundaresan (2021) suggests that indexical shift and perspective shift can be unified under a more general model of *attitude shift*, where various types of shifty elements are ordered in a single implicational continuum of attitude shift. According to this view, if indexical shift is possible in a linguistic environment in a language, perspective shift must also be available in the same environment.

In the current study we investigate the interactions between indexical shift and perspective shift from a different angle, testing if the two can (or must) occur at the same time in Turkish attitude reports, where indexical shift is optionally available (5).¹ We investigate how six different types of perspective sensitive items (PSIs) behave, and show that PSIs do not exhibit a uniform behavior and can be categorized into two main types depending on their interactions with indexical shift, as summarized in the table in (6).

- (5) Burak_i ben_{speaker/i} Ankara-ya git-ti-m san-ıyor.
 Burak 1SG Ankara-DAT go-PST-1SG think-PROG
 ‘Burak thinks that {I/he} went to Ankara.’

- (6) Shift together patterns between indexicals and different types of PSIs²

grayGroup	PSI type	Shift together	Relevant coordinate
Indexical PSIs	Motion verb <i>gel</i> ‘to come’ Evidentiality	YES	<i>author</i>
Non-indexical PSIs	Epistemic modals Predicates of personal taste Relative expressions Epithets	NO	<i>judge</i>

Some PSIs, which we term *Indexical PSIs*, show indexical-like properties and exhibit shift together patterns with the 1st person indexicals. In other words, Indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential markers) must shift when indexical shift takes place, and cannot shift when indexical shift does not take place. Other PSIs, which we call *non-indexical PSIs*, on the other hand, are not affected by the (non-)occurrence of indexical shift and exhibit their own

¹ Unless cited otherwise, all Turkish data in this paper comes from the judgments of the author, who is a native speaker of Turkish. Most examples have been surveyed among 20 other native speakers to be judged inside specifically created context descriptions (provided with examples in this paper). Except for three native speakers, who systematically different in their judgments with shifted vs. non-shifted readings, all native speakers confirmed their judgments for the data provided.

² Though the list of PSIs can be extended with more different types, the current study is limited to six different types of PSIs due to time and space constraints: (i) motion verbs, (ii) evidentiality, (iii) epistemic modals, (iv) epithets (v) predicates of personal taste, and (vi) socio-cultural expressions. Future research should extend this list to cover other types of PSIs.

patterns (e.g. optionally shifting, not being able to shift) regardless of indexical shift. The behaviors of these PSIs are not affected by occurring in an indexical shifting environment.

These results have important implications regarding the theoretical mechanisms of how PSIs are interpreted and what type of elements can be targeted by indexical shift. First, our findings show that PSIs are not uniform in how their interpretations are derived: Only a subset of PSIs (Indexical PSIs) are sensitive to the *author* coordinate, while most others are not directly sensitive to this parameter. This suggests PSIs might be sensitive to different context coordinates like *judge* and *author*. Second, our findings show for the first time that indexical shift can target not only nominal indexical elements (e.g. *I*, *you*) but also predicates (e.g. *gel* ‘to come’) and modals (evidential markers), suggesting that these elements can have indexical properties. This supports the context-overwriting approaches to indexical shift (e.g. Anand & Nevins 2004; Deal 2020), and are not directly predicted by approaches that reduce indexical shift into semantic/syntactic binding (e.g. Podobryaev 2014) or simple feature deletion/impoverishment operations (e.g. Messick 2023).

To account for the indexical-like properties of indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential marker ‘-mİş’), I propose that these PSIs have semantic definitions along the lines of (7), which makes them sensitive to the *author* coordinate.

- (7) a. *gel* = an event of movement towards the *author*
 b. ‘-mİş’ = the *author* lacks direct evidence for reported proposition

Assuming definitions along the lines of (7), I follow previous accounts of indexical shift in Turkish (Özyıldız et al. 2018; Akkuş 2019; Oğuz et al. 2020) and suggest that a context-overwriting shifty operator (OP_{AUTH}) can optionally merge the periphery of the finite embedded clauses in Turkish and replace the *author* coordinate of the context with the attitude holder, as schematized in (8).

- (8) Speaker = *Dursun*_{*i*}
 a. [TP Burak_{*k*} [CP/TP 1SG_{*i*} ... *come*_{*i*}/evidentials_{*i*}] think] non-shifted
 b. [TP Burak_{*k*} [CP/TP OP_{AUTH} [1SG_{*k*} ... *come*_{*k*}/evidentials_{*k*}]] think] shifted

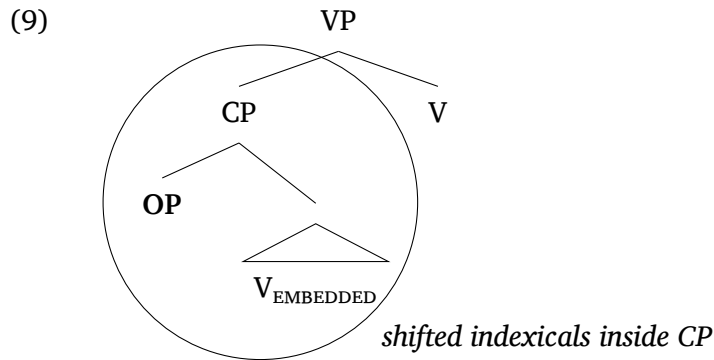
Once this OP_{AUTH} merges, it shifts the readings of all items that are sensitive to the *author* coordinate like 1st person pronouns and indexical PSIs.

The paper is organized as follows: Section 2 provides background on indexical shift and summarizes previous work on indexical shift in Turkish. Section 3 provides novel data demonstrating that some PSIs follow shift together patterns with indexicals while others do not, laying out the empirical generalizations. Section 4 lays out the analysis developed in this study, and Section 5 concludes the paper.

2 Background

One generally employed analysis of indexical shift is the shifty operator (or the monster operator) approach, originally proposed by Anand & Nevins (2004) and developed by subsequent work (e.g. Sudo 2012; Shklovsky & Sudo 2014; Deal 2020; but see Schlenker (1999, 2003) for a different approach). Under this view, a context-overwriting operator (OP) merges in the left periphery of the embedded clause (9) and forces all indexicals in its scope to be interpreted inside the reported context, rather than the actual speech context. For the Uyghur example in (2), this means that the OP merges in the left periphery of the embedded clause and the

1st person indexical refers to *Ahmet* and the 2nd person indexical refers to *Aygül*, who are the *author* and the *addressee* of the reported context, respectively.



Anand & Nevins (2004) report that all indexicals in an embedded clause must be interpreted relative to the same context (either shifted or non-shifted), introducing the well-known *Shift Together* constraint. Deal (2020) provides cross-linguistic data and shows that not all indexicals shift together, but the indexicals that depend on the same context coordinate (e.g. *author*, *addressee*, *location*) must shift together. She proposes that each context parameter can be shifted by a separate context-overwriting operator. For example, she argues that the OP_{AUTHOR} is responsible for shifting the *author* parameter of the context, while $OP_{ADDRESSEE}$ is responsible for shifting the addressee parameter. Considering that the interpretation of all 1st person indexicals depends on the *author* parameter, all 1st person indexicals must have shifted interpretations when they are under the scope of OP_{AUTHOR} . Crucially, under this view, the merge of OP_{AUTHOR} does not affect the interpretation of any other types of indexicals like 2nd person and *locatives*, since their interpretations are dependent on the *addressee* and *location* parameters of the context, respectively. Considering this, it is possible for a language to allow 1st person indexicals to have shifted interpretations, while disallowing shifted interpretations of 2nd person indexicals (such as *Slave*, as reported in Deal (2020)). To account for the cross-linguistic variation, Deal (2020) proposes that at least four different types of context-overwriting operators must exist: OP_{AUTHOR} for 1st person indexicals like *I*, $OP_{ADDRESSEE}$ for 2nd person indexicals like *you*, $OP_{LOCATIVE}$ for locative indexicals like *here*, and OP_{TIME} for temporal indexicals like *tomorrow*. In addition, Deal argues that those operators are in a structurally hierarchical relationship such that the merge of a structurally higher OP (e.g. $OP_{ADDRESSEE}$) requires the merge of a lower OP (e.g. OP_{AUTHOR}). This predicts that the shift of certain indexicals (e.g. 2nd person) require the shift of other indexicals (e.g. 1st person), leading to shift together patterns among different types of indexicals (see Deal, 2020 for details).

Turkish allows indexical shift in finite embedded clauses like (10), which has received some attention in the literature (Şener & Şener 2011; Özyıldız 2012; Özyıldız et al. 2018; Akkuş 2019; Korotkova 2019; Kuram 2020; Oğuz et al. 2020).

- (10) Burak_i (**ben**)_{speaker/i} Ankara-ya git-ti-m san-ıyor.
 Burak 1SG Ankara-DAT go-PST-1SG think-PROG
 ‘Burak thinks that {I/he} went to Ankara.’

The 1st person indexical *ben* in the embedded clause of (10) can be interpreted as referring to the speaker (non-shifted) or the attitude holder *Burak* (shifted). In Turkish, indexical shift is observed with various types of indexicals such as the 1st person *ben* ‘I’, 2nd person *sen* ‘you’, locative *burası* ‘here’, and temporal *yarın* ‘tomorrow’, embedded under attitude verbs like *de* ‘to say’, *san* ‘to think’, *iste* ‘to want’, and *bil* ‘to know’ (Özyıldız 2012; Oğuz et al. 2020).

Şener & Şener (2011) report that the availability of indexical shift in Turkish may depend on the phonological form of the indexical elements. For example, they suggest that shifted readings are available when the 1st person indexical is phonologically silent (*pro*), but are unavailable when the indexical is overt. They propose that only a sub-type of silent 1st person indexicals can shift, which they name *pro_{de se}*. They argue that this specific pronominal comes with a context-overwriting operator OP (in the sense of Anand & Nevins (2004)) attached in its nominal structure (approximately left periphery of the NP layer), and this OP makes the indexical interpreted inside the reported context, rather than the speech context. Özyıldız et al. (2018) and Akkuş (2019)³ on the other hand, follow the more traditional version of the shifty operator analysis of Anand & Nevins (2004), and assume that a context-overwriting operator OP can attach in the left periphery of the embedded clauses in Turkish, making indexical shift available. Oğuz et al. (2020) mainly build on top of these analyses but report that data regarding indexical shift is more complicated. Precisely, they show that different types of indexicals (e.g. 1st person and temporal) show different properties under different embedding verbs. For example, they report that the embedding verb *de* ‘to say’ forces a shifted reading for person indexicals (e.g. 1st person and 2nd person) in the embedded clause, while *san* ‘to think’ only allows, but does not require, shifted readings. Moreover, some indexicals like *bura* ‘here’ shift under the embedding verb *de* ‘to say’, while they do not shift under the embedding verb *san* ‘to think’. Examples in (11) (adapted from Özyıldız (2012)) show how *bura* ‘here’ behaves differently under the embedding verbs *say* ‘to say’ and *san* ‘to think’.

(11) If uttered by *Dursun* in *Los Angeles*:

- a. Ali İzmir-de [*bura*-nın ne-ler-i güzel] *de*-miş-ti?
 Ali İzmir-LOC here-GEN what-PL-3SG.POSS pretty say-DUB-PST
 ‘In İzmir, what did Ali say was pretty in {İzmir/Los Angeles}?’
- b. Ali İzmir-de [*bura*-nın ne-ler-i güzel] *san*-mış-tı?
 Ali İzmir-LOC here-GEN what-PL-3SG.POSS pretty think-DUB-PST
 ‘In İzmir, what did Ali think was pretty in {*İzmir/Los Angeles}?’

While (11a) with the embedding verb *de* ‘to say’ is ambiguous regarding how *bura* ‘here’ is interpreted (shifted or non-shifted), (11b) with the embedding verb *san* ‘to think’ is unambiguous and *bura* ‘here’ can only be interpreted as referring to the location where sentence is uttered (non-shifted). This shows that some indexicals like *bura* ‘here’ can shift under some embedding verb like *de* ‘to say’ but it cannot shift under the other embedding verbs like *san* ‘to think’.

To account for these facts, Oğuz et al. (2020) adopt Deal’s (2020) approach and suggest that specific types of OPs can merge to the left periphery of the finite embedded clauses (to be selected by embedding verbs) and lead to shifted interpretations of indexicals of their own type. For example, under their account, the embedding verb *de* ‘to say’ in (11a) can select for an embedded clause with an OP_{LOC} in its periphery, and as a result lead to shifted readings of locative indexicals like *bura* ‘here’, while the embedding verb *san* ‘to think’ cannot select for an embedded clause with the OP_{LOC} and thus the locative indexical *bura* ‘here’ cannot have shifted readings in (11b). However, *san* ‘to think’ can select for an embedded clause with an OP_{AUTHOR} in its periphery, and as a result lead to shifted readings of 1st person indexicals, as

³ Akkuş (2019) reports and analyzes a different dialect of Turkish, where violations of the *Shift Together* constraint of indexical shift is possible. He builds a more detailed analysis of indexical shift to account for these violation cases. None of our informants share the judgments reported in Akkuş (2019), so we do not provide details of his analysis in this paper.

in (10)⁴. As laid out in the previous section (8), this paper builds on this analysis and argues that indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential markers) are also targeted by the context shifting operator OP_{AUTHOR} .

2.1 Diagnosing indexical shift

Before moving forward, it is important to establish the fact that these are true instances of indexical shift, rather than direct quotation (e.g. Burak thinks, ‘I went to Ankara’). Previous work uses diagnostics like NPI licensing and *wh*-licensing to show that embedded material in sentences like (10) is not direct quotation (e.g. Özyıldız 2012; Akkuş 2019; Oğuz et al. 2020). Below, I demonstrate those diagnostics and show that examples used here are true instances of indexical shift.

In affirmative sentences, the Turkish NPI *kimse* ‘anyone’ must be licensed by a negative item that takes scope over it (e.g. Keleşir 2001). As in (12), the NPI *kimse* ‘anyone’ in the embedded object position can be licensed by embedded (12a) or matrix (12b) negation. Crucially, the sentences are ungrammatical in the absence of negation.

- (12) a. Ali [Burak **kimse-yi** gör-*(**me**)-di] san-ıyor.
 Ali Burak anyone-ACC see-NEG-PST think-PROG
 ‘Ali thinks that Burak didn’t see anyone.’
 b. Ali [Burak **kimse-yi** gör-dü] san-*(**mı**)-yor
 Ali Burak anyone-ACC see-PST think-NEG-PROG
 ‘Ali doesn’t think that Burak saw anyone.’

Considering that quotation creates an opaque domain, blocking the licensing of NPI by a negative item from outside as in (13), examples like (14) with shifted indexicals show that these are instances of indexical shift, but not direct quotation. If the embedded clause of (14) was a direct quotation, this would prevent the licensing of the NPI item *kimse* and lead to ungrammaticality, which is not the case.

- (13) *John did **not** say “I saw **anyone**”.

- (14) Burak_i (ben)_i **kimse-yi** gör-dü-m san-**mı**-yor.
 Burak 1SG anyone-ACC see-PST-1SG think-NEG-PROG
 ‘Burak doesn’t think that {I/he} saw anyone.’

Further evidence for that embedded material in those clauses are not direct quotation comes from *wh*-licensing. Considering that direct quotations are islands for *wh*-extractions, as in (15), sentences like (16) suggest that examples of indexical shift are not direct quotations.

- (15) *Who_i did John say “I saw *t_i*”.

- (16) Burak_i (ben)_i **kim-i** gör-dü-m san-ıyor?
 Burak 1SG who-ACC see-PST-1SG think-PROG
 ‘Who does Burak think that {I/he} saw?’

⁴ Oğuz et al. (2020) propose that some form-specific OPs exist in the language, targeting only silent 1st person indexical *pro*.1SG ($OP_{AUTH-pro}$) or emphatic 1st person *ben kendim* ($OP_{AUTH-emphatic}$). These details are not relevant for the purposes of the current study, and thus not discussed in further detail.

Turkish does not have overt *wh*-movement, so the example in (16) does not illustrate an overt extraction of the *wh*-element. However, the fact that the *wh*-item *kim* ‘who’ can get matrix scope interpretation suggests that it may covertly be extracted into the matrix domain, demonstrating that the embedded clause in (16) is not direct quotation.

So, the NPI licensing and *wh*-licensing facts discussed above show that we are dealing with true instances of indexical shift. We mostly use affirmative sentences in the rest of the paper for demonstration purposes, but NPI-licensing and *wh*-movement diagnostics were used to confirm the non-quotational status of each data point provided.

3 Perspective sensitive items inside *Indexical Shift* environments

This section provides novel data in order to investigate how Perspective Sensitive Items (PSIs) are interpreted inside indexical shifting environments. Novel data with PSIs such as the motion verb *gel* ‘to come’, evidential markers, epistemic modals (e.g. *may*, *might*), predicates of personal taste (e.g. *tasty*, *fun*), relative socio-cultural expressions (e.g. *local*, *foreigner*), and epithets (e.g. *idiot*, *bastard*) shows that the PSIs in Turkish do not exhibit a uniform behavior, and can be divided into two categories based on their interactions with the 1st person indexicals. Table in (17), repeated from (6), summarizes our findings.

(17) Shift together patterns between indexicals and different types of PSIs

Group	PSI type	Shift together	Relevant coordinate
Indexical PSIs	Motion verb <i>gel</i> ‘to come’ Evidentiality	YES	<i>author</i>
Non-indexical PSIs	Epistemic modals Predicates of personal taste Relative expressions Epithets	NO	<i>judge</i>

Crucially, *indexical PSIs* (e.g. motion verb *gel* ‘to come’, evidentials) behave similarly in that they show ‘shift together’ patterns with the 1st person indexicals and only shift when indexical shift takes place. *Non-indexical PSIs*, on the other hand, behave similarly in that they are not affected by the (non-)occurrence of indexical shift.

The following subsections provide relevant datapoints for each type of PSIs to demonstrate how they behave in indexical shifting environments.

3.1 Indexical PSIs: PSIs that must *shift together* with indexicals

This subsection introduces novel data focusing on the motion verb *gel* ‘to come’ and evidential markers. The data shows that the interpretations of the motion verb *gel* ‘to come’ and evidential markers are correlated with the interpretation of the 1st person indexical pronoun, showing shift together patterns. In other words, these PSIs behave like indexicals, and shift whenever indexical shift takes place. As shown in (7), I argue that these PSIs have semantic definitions that refer to the *author* coordinate of the context, and thus they are sensitive to indexical shift.

3.1.1 Motion verb *gel* ‘to come’ and *Indexical Shift*

Despite its complicated semantics and a wide range of uses, the motion verb *come* generally needs to be anchored to a perspective holder, describing a movement towards (the proximity

of) this perspective holder, who is typically *the speaker* of the sentence (e.g. Nakazawa 1990; Kubo 1997; Anderson 2021). For example, if *John* utters (18), it implies that *John* is in Los Angeles (i.e. *come* describes a movement towards the speaker *John*). An alternative reading is possible via *listener-anchoring* in a dialogue context, where the verb describes a movement towards the addressee of the current speaker.⁵ A sentence like (18) is typically considered infelicitous if neither the speaker *John* nor his addressee is in Los Angeles. This shows that the motion verb *come* in (18) must be anchored to the the speaker *John* (or his addressee), who is the default perspective holder of the sentence.

- (18) If *John* is speaking:
Andrew *came* to Los Angeles. → ‘John (or addressee) is in Los Angeles’

The interpretation of *come* can change under attitude reports, as in (19), which is a property generally attributed to perspective sensitive items (PSIs) (e.g. Oshima 2006; Bylinina et al. 2014).

- (19) Natasha said that Chomsky is *coming* to New Jersey next week.
a. → Speaker is in New Jersey (but Natasha need not be).
b. → Natasha is in New Jersey (but speaker need not be).
(from Sudo, 2018, ex. (40))

In (19), *come* can describe a movement towards the speaker (or the addressee) as the default perspective holder (19a), or it can undergo perspective shift and describe a movement towards the attitude holder *Natasha* (19b). When perspective shift takes place, the motion verb *come* is anchored to the perspective of the attitude holder *Natasha*, and thus *come* describes a movement towards *Natasha*, rather than the speaker.

Similar facts are observed in Turkish with the motion verb *gel* ‘to come’. If *Dursun* utters the simple sentence in (20), it implies that *Dursun* is in Ankara. Similar to English, listener-anchoring is possible in a dialogue situation, where *gel* ‘to come’ describes a movement towards the addressee.

- (20) If *Dursun* is speaking:
Burak Ankara-ya *gel*-di.
Burak Ankara-DAT come-PST
‘Burak came to Ankara.’ → ‘Dursun (or addressee) is in Ankara or its proximity.’

(20) is infelicitous if neither the speaker *Dursun* nor his addressee is in Ankara. This shows that *gel* ‘to come’, like its English counterpart, describes a movement towards the speaker as the default perspective holder. Crucially, *come* can be used to describe a movement to the proximity of the speaker (or addressee) from a further distance. For example, (20) is felicitous if either the speaker or the addressee is located in Konya, a city neighboring Ankara, if Ali goes to Ankara from a city further away (e.g. Istanbul).

Similar to English, *gel* ‘to come’ can undergo perspective shift in attitude reports, as shown in (21). Precisely, *gel* ‘to come’ in (21) can be interpreted as a movement towards the speaker *Dursun* (or the addressee) by default (21a), or it can be interpreted as a movement towards the attitude holder *Burak* as a result of perspective shift (21b). Finite embedded clauses like (21)

⁵ Though the motion verb *come* has a wide range of uses with complicated pragmatic functions, I concentrate here on the fact that in its simple uses *come* presupposes that there is a movement towards to perspective holder. For a more detailed discussion of *come* and its different uses, see Anderson (2021) and work cited in there.

are useful tools to investigate the interactions between *gel* ‘to come’ and indexicals, as both types of items can optionally shift when they appear inside those clauses.

- (21) If *Dursun* is speaking:
 Burak Ali Ankara-ya *gel*-di san-ıyor.
 Burak Ali Ankara-DAT come-PST think-PROG
 ‘Burak thinks that Ali came to Ankara.’
 a. → ‘Dursun (or addressee) is in Ankara or its proximity.’ (non-shifted)
 b. → ‘Burak is in Ankara or its proximity.’ (shifted)

Similar to English *come*, *gel* ‘to come’ has complex properties and a wide range of uses, which are outside the scope of this study. For example, *gel* ‘to come’ can be anchored to a different reference time than the actual utterance time, as shown in (22).

- (22) If uttered today, in Los Angeles:
 Geçen yaz Burak Tokyo-ya *gel*-di.
 last summer Burak Tokyo-DAT come-PST
 ‘Burak *come* to Tokyo last summer.’

(22) implies that the speaker or the addressee was in Tokyo last summer, and it is felicitous even though the speaker and the addressee are not in Tokyo at the time of the utterance. Assuming the speaker and the addressee are in Los Angeles, (22) would be infelicitous if neither the speaker nor addressee were in Tokyo last year. Examples like (22) show that reference time of *gel* ‘to come’ can be shifted, paralleling English *come* (e.g. Oshima 2006; Sudo 2018; Anderson 2021).

In addition, *gel* ‘to come’ can be anchored to the home-base of the speaker, and thus can generally be used to describe motion towards the home-base of the speaker.

- (23) If *Dursun* is from Turkey, and he is speaking in *Los Angeles*:
 Trump son 10 yıl-da hiç Ankara-ya *gel*-me-di.
 Trump last 10 year-LOC ever Ankara-DAT come-NEG-PST
 ‘Trump did not *come* to Ankara even once in the last 10 years.’

In (23), the speaker can use *gel* ‘to come’ to describe motion towards Ankara even though they are located in Los Angeles, because Ankara (or Turkey, in general) is the home-base of the speaker at the time of the utterance (e.g. the speaker may be visiting Los Angeles as a tourist).

Gel ‘to come’ is also used to mean ‘to arrive’ in certain contexts, where it appears to be more easily anchored to any salient entity. For example, if you ordered a gift for your friend Burak, who lives on the other side of the country, and you recently hear from Burak that the package was delivered, you can use *gel* to express that the gift has ‘arrived’ (24).

- (24) Speaker is in *Los Angeles*, *Burak* is in *Ithaca*:
 Burak-ın kargo-su *gel*-miş. Bence *pro*.3SG hediye-yi beğen-ecek.
 Burak-GEN package come-EVID. I.think gift-ACC like-FUT
 ‘Burak’s package has *come*/arrived. I think he will like the gift.’

In the current study, I focus on the simple cases with *gel* ‘to come’, when it is used to describe a movement towards the perspective holder, including the proximity of the speaker’s location.⁶ Though the other uses mentioned above are interesting, inclusion of those uses make the data extremely complicated, thus I do not focus on such uses.

Now that we know that *gel* ‘to come’ can shift inside finite embedded clauses, as in (21), the next step is to test its ability to shift is affected by indexical shift when it occurs in the same environment with an indexical. This can be tested in sentences like (25), which has the 1st person indexical *ben* as the embedded subject, and the motion verb *gel* ‘to come’ as the embedded verb. As shown in Section 2, the 1st person indexical *ben* can be interpreted as referring to the speaker (non-shifted) or to the attitude holder (shifted). Also, as we saw in (21), the motion verb *gel* can be interpreted as a movement towards the speaker or the speaker’s addressee (non-shifted), or as a movement towards the attitude holder (shifted). Considering that there are two elements that each can have *shifted* or *non-shifted* readings, (25) can in principle be ambiguous in four different readings if the indexical *ben* and motion verb *gel* can shift independently from each other. However, we see that this is not true for (25). Subscripts indicate who is in *Boston* or its proxy (e.g. New York), implied by the motion verb *gel* ‘to come’ (the destination of the movement denoted by *gel*).

- (25) If *Dursun* is speaking in *Los Angeles*:
 Burak [CP (*ben*) Boston-a *gel*-ey-im] ist-iyor.
 Burak 1SG Boston-DAT come-OPT-1SG want-PROG
 Lit: ‘Burak wants me to come to Boston’
- neither shifts → ‘Burak wants **speaker** to go to **Boston**_{#SPKR/ADDR}.’
 - both shift → ‘Burak wants **Burak** to go to **Boston**_{BURAK}.’
 - *only *gel* shifts → ‘Burak wants **speaker** to go to **Boston**_{BURAK}.’
 - *only *ben* shifts → ‘Burak wants **Burak** to go to **Boston**_{#SPKR/ADDR}.’

Out of the logically possible four interpretations, (25) can have only two. The first available reading is where neither the indexical *ben* nor the motion verb *gel* shifts (25a). In this reading, the indexical *ben* refers to the speaker, and *gel* describes a motion towards the addressee of the speaker (via listener-anchoring), which is visible with the implication that the addressee of the sentence should be in Boston⁷. This yields a reading like ‘The roommate wants the speaker to move towards the addressee’, and (25) is natural in this reading in a context like (26a). The second available reading is where both indexical *ben* and the motion verb *gel* shift (25b). In this reading, the indexical *ben* refers to the attitude holder *the roommate*, and *gel* also describes a motion towards *the roommate*, yielding to a reading like ‘The roommate wants the roommate to move towards the roommate’, which is odd/nonsensical unless it is anchored to a future time (i.e. roommate wants to go to their current location again in the future). (25) is natural in a context like (26b).

Importantly, both (25a) and (25b) require that the indexical *ben* and the PSI *gel* are get interpreted inside the same context: either the context of utterance (25a), or the reported context (25b). The readings where only one of these items shift (but not the other) are not available in the language (25c)-(25d), even in contexts/scenarios where those readings would be felicitous.

⁶ Proximity can be defined as a contextually determined spatial region anchored to the perspective holder (e.g., speaker, addressee, or shifted attitude holder), such that movement into this region counts as movement “towards” that individual for the purposes of licensing *gel* ‘come’.

⁷ *Gel* ‘to come’ is not interpreted as a movement towards the speaker here, since the target of the movement is explicitly *Boston*, and we know from the context that the speaker is in Los Angeles. The crucial observation here is that the motion verb describes a movement towards the participant of the speech act context (i.e. *addressee*), but not towards a referent from the embedded context (e.g. *attitude holder*). Also, since the attitude context does not have an addressee (as a thought), the listener anchoring is not available in shifted readings.

itous. For example, (25c) is not available even in a context like (26c), where ‘the roommate is in Boston and the roommate wants the speaker to go to Boston’, and (25d) is not available even in a context like (26d), where ‘the speaker is in Boston and the roommate wants to go to Boston’.

- (26)
- a. **Context A (neither shifts):** Dursun and Burak are in Los Angeles, and Dursun is on a phone call with his friend, who is in New York, which is in the proximity of Boston, relative to Los Angeles. Dursun tells his friend that Burak wants Dursun to go to Boston.
 - b. **Context B (both shift):** Dursun and his friend are having a conversation in Los Angeles. Burak is in New York, which is in the proximity of Boston, relative to Los Angeles. Dursun tells his friend that Burak wants to go to Boston (sometime in the future).
 - c. **Context C (only *gel* shifts):** Dursun and his friend are having a conversation in Los Angeles. Burak is in New York, which is in the proximity of Boston, relative to Los Angeles. Dursun tells his friend that Burak wants Dursun to go to Boston to visit him.
 - d. **Context D (only *ben* shifts):** Dursun and Burak are in Los Angeles, and Dursun is on a phone call with his friend, who is in New York, which is in the proximity of Boston, relative to Los Angeles. Dursun tells his friend that Burak wants to go to Boston for a conference.

These facts suggest the interpretation of the motion verb *gel* ‘to come’ is sensitive to indexical shift. Precisely, *gel* ‘to come’ must shift when it appears inside an indexical shift context, and it cannot shift without indexical shift. In other words, *gel* ‘to come’ and the 1st person indexicals must ‘shift together’. As I explain below, this shift together pattern follows naturally the analysis developed in this study. If *gel* ‘to come’ is interpreted relative to the *author* coordinate of the context, just like the 1st person indexical, it must shift whenever indexical shift (of the 1st person) takes place and it cannot shift without indexical shift.

3.1.2 Evidentiality and Indexical Shift

Evidentiality encodes the source of information or strength of evidence for a reported proposition (Aikhenvald 2004). While English uses various types of adverbs or modal verbs to encode evidentiality (e.g. *apparently*), other languages including (but not limited to) Bulgarian (Izvorski 1997), Cusco Quechua (Faller 2002), Korean (Park 2014), Finnish (Kaiser 2022b) St’at’imcets (Matthewson et al. 2007), and Turkish (Şener 2011) use specific evidential affixes or clitics. For example, Turkish uses evidential past tense morpheme ‘-miş’ to roughly express the lack of direct evidence for the described situation (e.g. Göksel & Kerslake 2005; Şener 2011; Meriçli 2016). The contrast between the regular past tense morpheme ‘-DI’ and the evidential past tense marker ‘-miş’ is demonstrated below.

- (27) If *Dursun* is speaking:
- a. Ali şarkı söyle-di.
Ali song sing-PST
‘Ali sang a song.’ → ‘Dursun has direct evidence for Ali’s singing’
 - b. Ali şarkı söyle-miş.
Ali song sing-EVID
‘Ali sang a song.’ → ‘Dursun lacks direct evidence for Ali’s singing’

Sentences (27a) and (27b) are minimally different in that the regular past tense marker ‘-DI’ is used in (27a) while the evidential past tense marker ‘-mİş’ is used in (27b). If uttered by *Dursun*, (27a) implies that *Dursun* has direct evidence for the reported event (e.g. he saw or heard Ali while Ali was singing). On the other hand, (27b) implies that *Dursun* lacks direct evidence for the reported event, and he received the information from an external source (e.g. someone else told *Dursun* that Ali sang a song) or inferred it from the context (e.g. he saw that Ali’s favorite microphone is outside its box). Considering that evidentiality is perspective sensitive, evaluated based on evidence available to the author of an utterance (Izvorski 1997; Speas & Tenny 2003; Stephenson 2007: a.o.), it is natural/expected that (27b) implies that the speaker, as the default perspective holder, lacks direct evidence for the reported situation.

Previous literature reports that evidentials can also undergo *perspective shift* in attitude reports and get evaluated under the perspective of the attitude holder (Sauerland & Schenner 2007; Stephenson 2007; Matthewson et al. 2007; Bylinina et al. 2014; Korotkova 2015; 2019; Kaiser 2024). This is observed in Turkish as well (Şener 2011; Meriçli 2016), as in (28), where the evidential marker ‘-mİş’ signals lack of direct evidence for the reported proposition.

- (28) Burak Ali öl-müş san-ıyor.
 Burak Ali die-EVID think-PROG
 ‘Burak thinks Ali apparently died.’

Importantly, (28) indicates that the attitude holder *Burak* lacks direct evidence for the reported situation, and (28) is felicitous even in contexts where the actual speaker has direct evidence for or against the reported situation.⁸

Establishing these facts, we can test sentences that have both indexical elements and evidential markers to check if they can/must shift together. Understanding the interactions between indexicals and evidentiality requires building up some context scenarios, as both of these items are highly context dependent. We can use a context like (29) and a sentence like (30) as a baseline.

- (29) Context 1: *Dursun takes a test and feels very confident that he will get a passing grade. The next day, his friend Pelin tells him that she talked to the professor and she knows that Dursun passed the test. After hearing this from Pelin, Dursun knows that he passed the test, as expected, even though he does not have direct evidence for it (i.e. he only has hear-say evidence).*

- (30) If *Dursun* is speaking:
 (Ben) sınav-ı geç-miş-im.
 1SG test-ACC pass-EVID-1SG
 ‘I apparently passed the test.’

Dursun can use (30) to describe the situation explained in Context 1. Importantly, *Dursun* in this example does not have direct evidence for that he passed the test. Thus, he can use the evidential marker ‘-mİş’ to report that he only has indirect (hear-say) evidence for his report.⁹ Sentences like (30) are important in that they contain an indexical (i.e. 1st person *ben*) and a perspective sensitive item (i.e. *evidential marker*) at the same time.

⁸ It is important to note that evidential marker can also be anchored to the listener in questions. For our purposes, the important observation is that it is by default interpreted in the actual context of utterance (either from the perspective of the actual *speaker* or the *addressee*).

⁹ In Turkish, ‘-mİş’ is also used as a mirative, indicating surprisal of the speaker for the reported proposition. To make sure that we focus on the evidential use of ‘-mİş’, we set up a context where the speaker expects to pass the test, so the mirative use of ‘-mİş’ is not available.

A sentence like (30) can be embedded under the attitude verb *san* ‘to think’ as in (31), where *indexical shift* is possible. Again, if the indexical 1st person and the evidential markers can shift independently, we predict that (31) should have four different logical interpretations. This prediction is not borne out. The subscripts next to ‘apparently’ indicate who lacks direct evidence for passing the test.

- (31) If *Dursun* is speaking:
 Burak (**ben**) sınav-ı geç-miş-im san-ıyor.
 Burak 1SG test-ACC pass-EVID-1SG think-PROG
 a. ‘Burak thinks that **speaker** apparently_{SPKR} passed the test.’
 b. ‘Burak thinks that **Burak** apparently_{BURAK} passed the test.’
 c. *‘Burak thinks that **speaker** apparently_{BURAK} passed the test.’
 d. *‘Burak thinks that **Burak** apparently_{SPKR} passed the test.’

The first available interpretation of (31) is where neither the 1st person indexical nor the evidential marker is shifted (31a). In this reading, the indexical refers to the speaker *Dursun*, and the evidentiality is also evaluated based on *Dursun*’s (speaker) perspective. This leads to a reading like ‘Dursun passed the test, and Dursun lacks direct evidence for his passing’. (31) could be used naturally in this reading in a context like (32a). The second available interpretation of (31) is where both the 1st person indexical and the evidential marker are shifted (31b). In this reading, the 1st person indexical refers to the attitude holder *Burak*, and the evidentiality is also evaluated based on *Burak*’s perspective. This leads to a reading like ‘Burak passed the test, and Burak lacks evidence for his passing’. (31) could be used naturally in this reading in a context like (32b), where Burak lacks, and the speaker has, direct evidence for that Burak passed the test.

- (32) a. **Context A (neither shifts):** Dursun takes a test and feels confident that he will get a passing grade. The next day, Pelin tells Dursun that she saw the grades in the professor’s office and he indeed got a passing grade. Dursun (who lacks direct evidence for his passing) shares the news with Burak, so Burak thinks that Dursun passed the test.
 b. **Context B (both shift):** Burak takes a test and feels confident that he will get a passing grade. The next day, Pelin tells Burak that she saw the grades in the professor’s office and he indeed got a passing grade. Burak (who lacks direct evidence for his passing) shares the news with Dursun. Unknown to Burak, it was Dursun who graded the tests, so Dursun already knew that Burak passed the test (Dursun has direct evidence).
 c. **Context C (only ‘-miş’ shifts):** Dursun takes a test and feels confident that he will get a passing grade. The next day, Pelin tells Burak that she saw the grades in the professor’s office and Dursun indeed got a passing grade. So, Burak believes that Dursun passed the test with no direct evidence. Unknown to Burak, the professor already showed his grade to Dursun in the morning (Dursun has direct evidence).
 d. **Context D (only *ben* shifts):** Burak takes a test and feels confident that he will get a passing grade. The next day, Pelin tells Dursun that she saw the grades in the professor’s office and Burak indeed got a passing grade. So, Dursun believes that Burak passed the test. Unknown to Dursun, the professor already showed his grade to Burak in the morning (Burak has direct evidence).

Similar to our observation with the motion verb *gel* ‘to come’ in the previous subsection, we see here that in both available interpretations the indexical *ben* and the evidential marker ‘-miş’ are interpreted relative to the same context: either the context of utterance (31a), or the

reported context (31b). Readings where only one of these two elements shifts are unavailable (31c)-(31d), even in contexts where those readings would be expected to be felicitous. For example, (31c) is not available even in a context like (32c), where ‘Burak lacks direct evidence for that the speaker passed the test’, and (31d) is not available even in a context like (32d), where ‘the speaker lacks direct evidence for that Burak passed the test’. This is due to the fact that these interpretations would force the 1st person indexical and the evidential marker to be interpreted in different contexts (i.e. reported context vs. actual speech context), which is not allowed.

In sum, echoing what we saw with the motion verb *gel* ‘to come’, the interpretation of evidential markers is correlated with *indexical shift* in Turkish finite embedded clauses. When indexicals shift, evidential markers must also shift, and when indexicals stay non-shifted, evidential markers cannot shift. In other words, the evidential markers and the 1st person indexicals must ‘shift together’. This, again, follows naturally under the analysis developed in this paper. If evidentiality is evaluated relative to the world knowledge (amount of evidence) of the *author*, it is expected to shift whenever indexical shift (of the 1st person) replaces the the *author* with the attitude holder, and is expected to stay non-shifted in the absence of indexical shift.

Data introduced here shows that the motion verb *gel* ‘to come’ and evidential markers both exhibit shift together patterns with the 1st person indexical and form the first sub-type of PSIs, which I call *Indexical PSIs*. As explained in more detail below, I suggest that Indexical PSIs have semantic definitions that refer to the speech-act coordinates of the context (e.g. *author*), as in (7), which makes them targeted by the indexical shifting operator. If semantic definitions along the lines of those are adopted, the behavior of indexical PSIs can be accounted for with the standard, OP-based analyses of indexical shift (e.g. Anand & Nevins 2004; Deal 2020).

3.2 Non-indexical PSIs: PSIs that do not shift together with indexicals

Our investigation of various types of PSIs reveals that not all PSIs show indexical-like properties. In this section, we provide data with epistemic modals, predicates of personal taste, relative socio-cultural expressions, and epithets inside indexical shifting contexts and show that they behave differently from indexical PSIs. Though these PSIs are not typically considered to form a uniform group (i.e. they have different properties within Turkish and cross-linguistically), the relevant observation for the purposes of this study is that they are all insensitive to indexical shift, which implies that they are not interpreted relative to the indexical *author* parameter of the context. Thus, we call them *non-indexical PSIs*.

One type of non-indexical PSIs is relative socio-cultural expressions such as *local*, *foreigner*, *enemy*, and *fellow citizen*, which are interpreted based on the socio-cultural status of a perspective holder (Mitchell 1986; Partee 1989; Oshima 2006; Bylinina et al. 2014). For example, a word like *foreigner* means something like ‘not from the same country with the perspective holder’. The perspective holder of the relative socio-cultural expressions (henceforth, relative expressions) in root clauses is typically the speaker by default, but they can undergo perspective shift and be interpreted from the perspective of another person.

Sentences like (33) allow us to test any possible interactions between relative expressions and indexical shift. (33) contains both the 1st person indexical *ben* and the relative expression *yerel* ‘local’ in its embedded clause. Considering that both of these items can optionally shift, a sentence like (33) has four different logically possible readings. If indexical shift and the perspective shift of relative expressions are independent from each other, we predict that the sentence should be ambiguous in four different readings. However, if indexical shift and perspective shift interact and they must (or cannot) take place together, the sentence should not be ambiguous in more than two different readings, as we observe with indexical PSIs.

- (33) If *Dursun_i* is speaking:
 Ken (ben) yerel bir gazete-ye röportaj ver-di-m san-ıyor.
 Ken 1SG local a journal-DAT interview give-PST-1SG think-PROG
 ‘Ken thinks that I/he gave an interview to a local journal.’
 a. → Ken thinks *Dursun* interviewed a Turkish journal (local to *Dursun*).
 b. → Ken thinks *Ken* interviewed a Japanese journal (local to *Ken*).
 c. → Ken thinks *Dursun* interviewed a Japanese journal (local to *Ken*).
 d. → Ken thinks *Ken* interviewed a Turkish journal (local to *Dursun*).

We see that all four readings of (33) are available, revealing that indexical shift and perspective shift can occur independently. More precisely, we see that both items can stay non-shifted as in (33a) (interpreted from the perspective of the speaker), they can both shift at the same time as in (33b) (interpreted from the perspective of the attitude holder), or one of them may shift while the other does not, as in (33c) and (33d). This shows that indexical shift and perspective shift of the relative expression *yerel* ‘local’ are independent from each other, and one can take place even if the other does not. This implies that the interpretation of some PSIs like relative expressions is not sensitive to the indexical coordinates of the context, and that PSIs should be divided into two categories as *indexical PSIs* and *non-indexical PSIs*.

Further evidence for the distinction of indexical shift and perspective shift of PSIs comes from examples with epistemic modals. The semantics of epistemic modals has been largely studied in the literature. Despite disagreements in the semantic details of how epistemic modals are derived, it has been reported that epistemic modals can have shifted or non-shifted readings inside attitude reports (e.g. Egan et al. 2005; Stephenson 2007; Yanovich 2014; Roberts 2023; Rudin 2024). Though the current study does not aim to provide any detailed analysis of how epistemic modals are derived in Turkish, our preliminary investigation suggests that epistemic modals in Turkish are by default interpreted from the perspective of the speaker in matrix clauses and from the perspective of the attitude holder (at least in stand alone sentences without rich contextual information). This is demonstrated for Turkish in examples (34a) and (34b).

- (34) a. Salı gün-ü yağmur yağ-abil-ir.
 Tuesday day-3SG.POSS rain fall-MOD-AOR
 ‘It may rain on Tuesday.’
 → It is compatible with *speaker*’s knowledge that it is possible that it rains on Tuesday.
 b. Burak Salı gün-ü yağmur yağ-abil-ir san-ıyor.
 Burak Tuesday day-3SG.POSS rain fall-MOD-AOR think-PROG
 ‘Burak thinks that it may rain on Tuesday.’
 → It is compatible with *Burak*’s knowledge that it is possible that it rains on Tuesday.

While the modal in (34a) is interpreted relative to the knowledge of the speaker, the modal in (34b) undergoes perspective shift and gets interpreted relative to the knowledge of the attitude holder *Burak*. Native speakers report that non-shifted readings are not available (at least easily) in (34b), suggesting that the default interpretation of epistemic modals shift in Turkish attitude reports is from the perspective of the attitude holder.

In light of this, sentences like (35), with a 1st person indexical and an epistemic modal in the embedded clause, show that the perspective shift of the modals and indexical shift are independent from each other.

- (35) If *Dursun* is speaking:
 Burak (**ben**) hasta ol-**abil**-ir-im san-ıyor.
 Burak 1SG sick be-MOD-AOR-1SG think-PROG
 a. ‘Burak thinks that I may be sick.’
 → It is compatible with *Burak*’s knowledge that it is possible that *Dursun* is sick.
 b. ‘Burak thinks that he may be sick.’
 → It is compatible with *Burak*’s knowledge that it is possible that *Burak* is sick.

As in (34b), we see that the modal in (35) shifts by default and is interpreted under the perspective of the attitude holder *Burak*. However, the indexical can stay non-shifted and refer to the speaker *Dursun* (35a), or it can shift and refer to the attitude holder *Burak* (35b). In other words, the indexical can stay non-shifted when the epistemic modal shifts, suggesting that perspective shift of modals and indexical shift are independent mechanisms. This contrasts recent work by Roberts (2023), who suggests that epistemic modals could be indexical.

A similar pattern is observed with predicates of personal taste (PPTs) as in (36), where a (dropped) 1st person possessive pronoun and the PPT *lezzetli* ‘tasty’ appear inside the same embedded clause.

- (36) If *Dursun_i* is speaking:
 Burak **pro**.1SG anne-m-in suşi-si çok lezzetli san-ıyor.
 Burak mother-1SG.POSS-GEN sushi-3SG.POSS very tasty think-PROG
 a. ‘Burak thinks that my mother’s sushi is very tasty.’
 → Sushi is judged to be very tasty by *Burak* (attitude holder).
 b. ‘Burak thinks that his mother’s sushi is very tasty.’
 → Sushi is judged to be very tasty by *Burak* (attitude holder).

Though the PPT shifts and is interpreted from the perspective of the attitude holder *Burak* by default¹⁰, the 1st person possessor can stay non-shifted (36a), referring to the speaker, or it can shift and refer to the attitude holder *Burak* (36b). So, again, the 1st person indexical can stay non-shifted even when the PPT undergoes perspective shift (36a), suggesting that the two phenomena are independent from each other.

Another piece of evidence for the separation of indexical shift and perspective shift of some PSIs comes from epithets. Epithets such as *the idiot* and *the bastard* express an evaluation or judgment towards their referents, and must be interpreted under the perspective of a judge/evaluator (e.g. Kratzer 1999; Speas & Tenny 2003; Potts 2004; Amaral et al. 2007; Harris & Potts 2009). Our investigation suggests that epithets in Turkish cannot shift inside attitude reports (at least, cannot shift easily¹¹), meaning that a speaker cannot use an epithet unless they share the judgment/evaluation towards its referent.¹² For example, the use of the epithet

¹⁰ As discussed in Stephenson (2007) and subsequent work, predicates of personal taste (PPTs) in attitude reports can be interpreted from the perspective of a judge other than the attitude holder, as in sentences like ‘My wife thinks that this cat food is tasty’, where *tasty* is naturally interpreted as ‘tasty for the cat’. Parallel examples exist in Turkish as well. The relevant fact here is that PPTs are generally interpreted from the perspective of the attitude holder in standalone sentences without specific semantics/pragmatics.

¹¹ Previous theoretical and experimental work suggests that epithets are by default interpreted as expressing the evaluations of the speaker, but the interpretation can shift in exceptional scenarios that require very rich contextual set ups (e.g. Harris & Potts 2009; Patel-Grosz 2012; Kaiser 2015).

¹² Some speakers note that they might take sentences like (37) to possibly express the judgment from the attitude holder’s perspective, if the speaker does not have any particular judgment towards *Ali* (e.g. they do not know *Ali* or they barely know anything about him). However, if they know *Ali* and disagree with the judgment that he is a scumbag, they say they cannot utter (37). For those speakers, including the author of the current study, it might be the case that they allow perspective shift of an epithet in specific contexts (e.g. when they do not have a judgment of the referent of the epithet), in line with the behavior reported for English (e.g. Harris & Potts 2009; Patel-Grosz

şerefsiz ‘scumbag’ in (37) indicates that the speaker (but not necessarily the attitude holder) thinks *Ali* is a scumbag. However, this does not seem to interact with indexical shift, as the 1st person indexical in the sentence can stay non-shifted (37a) or shift (37b).

- (37) If *Dursun* is speaking:
 Burak (ben) Ali şerefsiz-i-ni gör-dü-m san-ıyor.
 Burak 1SG Ali scumbag-3.POSS-ACC see-PST-1SG think-PROG
 a. ‘Burak thinks that I saw that scumbag Ali.’
 → *Dursun* (speaker) thinks Ali is a scumbag (non-shifted).
 b. ‘Burak_i thinks that he_i saw that scumbag Ali.’
 → *Dursun* (speaker) thinks Ali is a scumbag (non-shifted).

The fact that the indexical can shift while the epithet does not, as in (37b), suggests that the perspective shift of epithets and indexical shift are two independent mechanisms in Turkish.

In overall, we see that the shifting patterns of some PSIs like epistemic modals, predicates of personal taste, relative socio-cultural expressions, and epithets are not affected by the (non-)occurrence of indexical shift. As we have seen through examples (33)-(37), the interpretations of these PSIs do not change when indexical shift occurs. This suggests that these PSIs do not have indexical-like properties, unlike *indexical PSIs*, thus we categorize them as *non-indexical PSIs*.

In the next subsection I provide a summary of the patterns observed with different types of PSIs and develop an analysis to account for those patterns.

3.3 Summary of empirical picture

In summary, in the preceding sections I presented new empirical data demonstrating that perspective sensitive items (PSIs) do not exhibit a uniform behavior regarding their sensitivity to indexical shift. Data showed that the motion verb *gel* ‘to come’ (25) and evidential markers (31) *must* shift whenever they appear in an environment where indexical shifting takes place, and they *cannot* shift if indexical shift does not occur. This suggests that the interpretation of the motion verb *gel* ‘to come’ and evidential markers are sensitive to indexical shift, and that they must *shift together* with the 1st person indexicals. Since the interpretations of these PSIs are tied together with indexicals, we call them *indexical PSIs*. Other PSIs, which I call *non-indexical PSIs*, on the other hand, do not seem to be affected by whether indexical shift takes place or not. For example, relative socio-cultural expressions (e.g. *local*) can freely shift independently from indexical shift (33), epistemic modals (e.g. *may/might*) shift by default even in cases where indexical shift does not take place (35), and epithets (e.g. *idiot*) shift even when they appear in an indexical shift environment (37).

4 Proposal

The existence of *indexical PSIs* is surprising, considering that PSIs are traditionally treated as a different class of items than indexicals, and have been proposed to be interpreted relative to non-indexical coordinates like *judge* (Lasersohn 2005; Potts 2007; Amaral et al. 2007; Bylina et al. 2014; Kaiser 2022a: a.o). For example, Lasersohn (2005) suggests that predicates of personal taste (PPTs) are interpreted relative to the *judge* coordinate of the context, which

2012; Kaiser 2015). Most speakers, however, provided consistent judgments indicating that they do not allow perspective shift of an epithet.

represents the individual whose perspective the PPT is ‘judged’ from. Stephenson (2007) extends this analysis with modifications to epistemic modals (but see Kratzer 1977 and Yalcin 2007 for alternative accounts), and Potts (2007) extends it to epithets. Though the specific details of these accounts vary, the important matter relevant to the current study is that those accounts all suggest that PSIs like PPTs, epistemic modals, and epithets are not interpreted relative to the *author* coordinate of the context. Thus, according to this view, the interpretations of indexicals and PSIs should be independent from each other, and shift in one domain (e.g. indexical shift) should not force or block shift in the other domain (e.g. perspective shift). Though an approach like this is compatible with the patterns observed with some PSIs (e.g. PPTs, epithets), the patterns observed in Turkish with *indexical PSIs* suggest that at least some types of PSIs are interpreted in a similar way to indexicals, that is, they are sensitive to the indexical coordinates of the context. Based on this, I suggest that indexical PSIs in Turkish (e.g. motion verb *gel* ‘to come’ and evidential markers) are interpreted relative to the *author* coordinate of the context, rather than *judge*, and thus they must shift when they appear in an indexical shifting environment. This is in line with the predictions of some of previous work, which suggested that PSIs might possibly have indexical indexical properties and at least some types of perspective shift could be analyzed as indexical shift (e.g. Amaral et al. 2007; Percus 2011). However, this idea was abandoned mostly based on English data like (38), where PSIs can shift even though indexical shift is not available.

- (38) My father screamed that he would never allow **me** to marry that **bastard** Webster.
(from Kratzer, 1999, ex (14))

The famous example in (38), originally from Kratzer (1999), discussed in Amaral et al. (2007), shows that the 1st person indexical *me* refers to the speaker (non-shifted), while the epithet *bastard* shifts and gets interpreted from the perspective of the attitude holder ‘father’ (e.g. the speaker does not think Webster is a bastard). Considering that English does not allow indexical shift (as evident by the non-shifted *me*), (38) implies that the perspective shift of PSIs (e.g. *bastard*) is not due to indexical shift and it must be independent from indexical shift.

In a highly relevant study, Sudo (2018) takes a different approach and suggests that English allows indexical shift in presuppositional context, and some PSIs (e.g. *come*) can have shifted interpretations as a result of this *presuppositional indexical shift*. Sudo (2018) reports data in (39), demonstrating that English *come* can have shifted interpretations in attitude reports. More precisely, *come* in (39) can either imply that the speaker is in New Jersey (non-shifted), or that the attitude holder *Natasha* is in New Jersey (shifted).

- (39) Natasha said that Chomsky is *coming* to New Jersey next week.
a. → Speaker is in New Jersey (but Natasha need not be).
b. → Natasha is in New Jersey (but speaker need not be).
(from Sudo, 2018, ex. (40))

Sudo (2018) proposes that English *come* has an ‘indexical presupposition’ that needs to be satisfied. In simple terms, Sudo argues that *come* presupposes that there is a moving event towards the location of the *author* (speaker), and that the verb *come* can only be used to describe a moving event when this presupposition is satisfied. Whenever this presupposition is not satisfied (e.g. movement is not towards the *author*), the alternative motion verb *go* should be used instead.

In (39b), the motion verb *come* describes a movement towards the attitude holder (Natasha), rather than the speaker, suggesting that the *author* of the context in which *come* is interpreted is shifted (to Natasha). Considering that English does not allow indexical shift, Sudo argues

sentences with *come* are interpreted relative to two separate contexts (c_1 and c_2), and that presuppositional content are interpreted relative to the second context (c_2), as formalized in (40). According to Sudo, (39b) is an instance of ‘presuppositional’ indexical shift, where the *author* of c_2 (the context that is relevant for presuppositional content) is *Natasha*, rather than the actual speaker.

(40) Formalizing (39) according to Sudo (2018), simplified and ignoring tense:

$$(\text{Chomsky comes to New Jersey})^{g, c_1, c_2} = \lambda k. \exists x \in \text{Author}(c_2) [\text{Loc}_{w_{c_2}}(x, \text{New Jersey}, t_{c_2})]$$

According to this analysis, natural languages can possess two different types of monstrous operators. The first one (OP_1) is the standard indexical shifting operator that targets c_1 and shifts the interpretations of indexicals (e.g. *I*, *you*, *here*) in languages that allow indexical shift (e.g. Uyghur, Zazaki, Turkish). The second monstrous operator (OP_2) targets c_2 and shifts the context relative to which presuppositional content is evaluated, such as the presuppositional content of *come* in English.

This extra mechanism that Sudo develops is compatible with (but not required for) the Turkish data introduced here. In Turkish data provided here, we observe that what is described as presuppositional indexical shift (e.g. *gel* ‘to come’) must occur together with regular indexical shift (e.g. 1st person). Considering this, we could argue that the standard indexical shift (targeting c_1) and the presuppositional indexical shift (targeting c_2) must occur together in Turkish, such that the shifty operator responsible for the standard indexical shift hierarchically contains the operator responsible for the presuppositional indexical shift. Crucially, this would predict that all languages that allow the standard indexical shift (e.g. Turkish, Uyghur) would require presuppositional indexical shift to occur whenever indexical shift takes place (e.g. as we observe in Turkish). Alternatively, if we assume that (at least in Turkish) *gel* ‘to come’ is sensitive to indexical coordinates in its assertive content, rather than its presuppositional content, we could argue that only one shifty operator (i.e. responsible for the standard indexical shift in c_1) can derive the shift together facts observed in Turkish.

In sum, if one follows Sudo and assumes that verbs like *come* are interpreted relative to two separate contexts (i.e. c_1 and c_2), and that those verbs are indexical in their presuppositional content as in (40), one could argue that two separate indexical shift operations (standard indexical shift and presuppositional indexical shift) are responsible for the shift together facts observed in Turkish data introduced here. This approach requires the addition of a second shifty (monstrous) operator into the standard context-overwriting approaches to indexical shift (e.g. Anand & Nevins 2004; Deal 2020)), which is responsible for shift in presuppositional content, and a restriction such that in languages like Turkish this new operator must co-occur with the standard indexical shift operator. On the other hand, if one assumes that verbs like *gel* ‘to come’ are (at least in Turkish) indexical in their assertive content, the standard context-overwriting approaches can account for the facts laid out here without any modifications. In this approach, the shift of *gel* ‘to come’ is not in the presuppositional level, but rather in the assertive level similar to other indexical items (e.g. *I*, *you*, *here*).

Currently, there is no reliable diagnostic to test if the shift observed with *gel* ‘to come’ is presuppositional or standard indexical shift. As a result, both approaches can successfully account for the empirical data from Turkish. Considering that Sudo’s account requires a substantial modification to the standard shifty-context mechanisms (i.e. the addition of a second context parameter to the derivation), and requires a certain semantic definition of *come* (i.e. indexical properties in the presuppositional content), I simply follow the simpler, standard accounts, and assume that the standard (not presuppositional) indexical shift is responsible for the shifty

behavior of *gel* ‘to come’ in Turkish. The technical details of the semantics of motion verbs is out of the scope of this paper, so I do not provide a detailed formal discussion of its semantics. In broad terms, any semantic denotation that ties the interpretation of the motion verb *gel* ‘to come’ to the *author* parameter of the context (as Sudo does) could account for the shift together facts of *gel* and 1st person indexicals in Turkish. I leave the semantic details for future work.

A similar mechanism should be at play for evidential markers, where evidentiality is interpreted with reference to an indexical context parameter like *author*. There is a large literature suggesting that evidentiality is evaluated/interpreted based on the world knowledge of the speaker a sentence (Izvorski 1997; Speas & Tenny 2003; Sauerland & Schenner 2007; Stephenson 2007: a.o.). For example, Izvorski (1997) suggests that the use of an indirect evidential marker asserts that the reported proposition *p* is compatible with *the speaker’s* knowledge, and it presupposes that *the speaker* has indirect evidence for *p*. This is similar to Sudo’s proposal for *come*, discussed above, in that the interpretation of the evidential marker refers to the context coordinate *author*, which is sensitive to indexical shift. Though there are different views on the details of the semantics of evidentiality, any approach that ties the interpretation of evidential markers to the indexical coordinate *author* should successfully account for the shift together facts of evidentials and 1st person indexicals in Turkish.

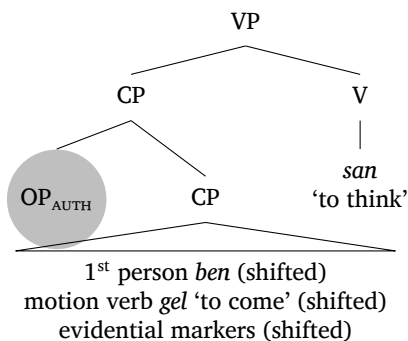
Leaving the details of their semantics for future work, I propose that the motion verb *gel* ‘to come’ and the evidential markers like ‘*-mİş*’ in Turkish are sensitive to the indexical coordinate *author*. I suggest that the semantics of *gel* and the evidential marker ‘*-mİş*’ are along the lines of (41a) and (41b).¹³

- (41) a. *gel* = an event of movement towards the *author*
 b. ‘*-mİş*’ = the *author* lacks direct evidence for reported proposition

If semantic descriptions along the lines of (41) are adopted, the shifting (and shift together) facts of indexical PSIs can be straightforwardly captured under the standard context-overwriting accounts of indexical shift (e.g. Anand & Nevins 2004; Deal 2020). I follow previous work in Turkish (Özyıldız et al. 2018; Akkuş 2019; Oğuz et al. 2020) and propose that a context overwriting operator OP_{AUTHOR} can merge in the left periphery of the finite embedded clauses in Turkish and replace the *author* parameter of the context with the attitude holder, schematized in (42) and (43).

- (42) Speaker = *Dursun_i*
 a. [TP Burak_k [CP/TP 1SG_i ... *come_i/evidentials_i*] think] non-shifted
 b. [TP Burak_k [CP/TP OP_{AUTH} [1SG_k ... *come_k/evidentials_k*]] think] shifted

¹³ These are descriptive explanations of a semantic analysis that would be compatible with the Turkish data introduced in this paper, and are not intended as formal proposals. I leave the semantic/formal details of these items for future work.

(43) *Shifty OP embedded under san ‘to think’*

The merge of this OP_{AUTH} changes the context under which indexicals (1st person) and indexical PSIs (*gel* ‘to come’ or evidential markers) are interpreted, replacing the value of the *author* coordinate with the attitude holder (42b). In the absence of this OP_{AUTH} , all of these items are interpreted in the actual speech context, where the speaker is the *author* (42a). Assuming that the semantics of *gel* and ‘-*mlş*’ are roughly as in (41a) and (41b), the shift together facts observed in sentences like (25) and (31), repeated here as (44) and (45), are expected since the merge of OP_{AUTH} changes the context that the indexicals and indexical PSIs are interpreted in, forcing both to shift.

(44) *If Dursun is speaking in Los Angeles:*

Burak [CP (OP_{AUTH}) (*ben*) Boston-a *gel*-ey-im] ist-iyor.
 Burak 1SG Boston-DAT come-OPT-1SG want-PROG
 Lit: ‘Burak wants me to come to Boston’

- a. neither shifts → ‘Burak wants **speaker** to go to **Boston**_{#SPKR/ADDR}.’
- b. both shift → ‘Burak wants **Burak** to go to **Boston**_{BURAK}.’
- c. *only *gel* shifts → ‘Burak wants **speaker** to go to **Boston**_{BURAK}.’
- d. *only *ben* shifts → ‘Burak wants **Burak** to go to **Boston**_{#SPKR/ADDR}.’

In (44), the indexical shifting operator OP_{AUTH} can merge in the left periphery of the embedded clause and shift the *author* parameter of the context in which the sentence is interpreted. As a result, all indexical elements that are sensitive to the *author* parameter, 1st person *ben* and *gel* ‘to come’ in this case, will be interpreted in a context where the roommate is the *author*, leading to the interpretation in (44b). When this OP_{AUTH} is absent, the sentence is interpreted in the actual speech context, where both 1st person *ben* and *gel* ‘to come’ are interpreted as referring to the speaker *Dursun* as the *author*, and the addressee is available for listener-anchoring (44a). Crucially, since both 1st person *ben* and *gel* ‘to come’ depend on the *author* parameter of the context, they can either shift together or they cannot shift at all, so the readings where one shifts without the other are not available (44c)-(44d).

Similar to (44), a context-overwriting OP_{AUTH} can optionally merge in the embedded clause of (45), and replace the *author* parameter of the context in which the sentence is interpreted with the attitude holder *Burak*. This forces the 1st person indexical and the evidential marker ‘-*mlş*’ to be interpreted in a context where *Burak* is the *author*, and leads to the shifted interpretation in (45b). In the absence of the OP_{AUTH} , the sentence is interpreted in the actual speech context, where the actual speaker *Dursun* is the *author*, leading to the interpretation in (45a). Again, since both 1st person *ben* and ‘-*mlş*’ depend on the same context coordinate, they can either shift together or not shift at all, and the readings where only one shifts without the other are not available (45c)-(45d).

- (45) If *Dursun* is speaking:
 Burak (**ben**) sınav-ı geç-miş-im san-ıyor.
 Burak 1SG test-ACC pass-EVID-1SG think-prog
 a. ‘Burak thinks that **speaker apparently**_{SPKR} passed the test.’
 b. ‘Burak thinks that **Burak apparently**_{BURAK} passed the test.’
 c. *‘Burak thinks that **speaker apparently**_{BURAK} passed the test.’
 d. *‘Burak thinks that **Burak apparently**_{SPKR} passed the test.’

Importantly, the analysis developed here suggests that the shifted readings of indexical PSIs (e.g. the motion verb *gel* ‘to come’ and evidential markers) are due to the occurrence of indexical shift, rather than perspective shift (of the *judge*), which allows shifted readings for other types of PSIs. As a result, this approach predicts that indexical PSIs cannot have shifted readings in environments where indexical shift is not available. This prediction is borne out, and can be demonstrated by examples with nominalized embedded clauses. It has been well documented that nominalized embedded clauses like (46) in Turkish do not allow indexical shift (e.g. Şener 2011; Özyıldız 2012). In (46), the 1st person embedded subject must be interpreted as referring to the speaker, and cannot shift and refer to the matrix subject *Ali*.

- (46) Ali (**ben-im**) Boston-a git-tiğ-im-i san-ıyor.
 Ali 1SG-GEN Boston-DAT go-NMLZ-1SG-ACC think-PROG
 ‘Ali thinks that {I/*he} went to Boston.’

Crucially, even though indexical shift is not allowed in nominalized clauses, (non-indexical) PSIs can still undergo perspective shift. This is demonstrated in (47), with the relative socio-cultural expression *yerel* ‘local’. Assuming that *Dursun* is Turkish, *Ken* is Japanese, and *John* is American, we see that the relative expression *yerel* ‘local’ can be anchored to the speaker (Turkish), to the embedded subject (Japanese), or to the matrix subject (American).

- (47) If *Dursun* is speaking:
 John Ken-in yerel bir gazete-ye röportaj ver-diğ-i-ni san-ıyor.
 John Ken-GEN local a journal-DAT interview give-NMLZ-3SG-ACC think-PROG
 ‘John thinks that Ken interviewed a local_{TURKISH/JAPANESE/AMERICAN} journal.’

Examples like (46) and (47) show that indexical shift is not allowed in nominalized embedded clauses, while perspective shift (of PSIs) is allowed. So, if the motion verb *gel* ‘to come’ behaves like other PSIs, we would expect it to be able to shift inside nominalized embedded clauses. However, if *gel* behaves like indexicals (and its shift is due to this property, as proposed here), we predict it to be unable to shift inside nominalized clauses. This prediction is borne out, as demonstrated in (48).

- (48) a. Ali (**ben-im**) Boston-a gel-diğ-im-i san-ıyor.
 Ali 1SG-GEN Boston-DAT go-NMLZ-1SG-ACC think-PROG
 ‘Ali thinks that {I/*he} came to Boston.’ → ‘Spkr (or Addr) is in Boston’
 b. Ali Ayşe-nin Boston-a gel-diğ-in-i san-ıyor.
 Ali Ayşe-GEN Boston-DAT go-NMLZ-3SG-ACC think-PROG
 ‘Ali thinks that Ayşe came to Boston.’ → ‘Spkr (or Addr) is in Boston’

Sentences in (48) have the motion verb *gel* ‘to come’ inside nominalized embedded clauses. Both sentences in (48) imply that the speaker (or addressee, via listener-anchoring) should be in Boston, and they cannot be uttered in a context where neither the speaker nor the addressee

is in Boston. Importantly, these sentences do not have any implications regarding the location of the matrix subject *Ali*, suggesting that *gel* cannot shift in this environment. This is different from finite embedded clauses (44), where indexical shift is available. Here, we see that the unavailability of indexical shift blocks shifted readings of *gel* ‘to come’, with (48a) or without (48b) an indexical element in the same domain. This provides further evidence for the analysis developed here, suggesting that *gel* ‘to come’ can shift due to its indexical properties, only when it appears in an indexical shifting environment.¹⁴

Lastly, it is worth mentioning that the existence of indexical PSIs suggests that elements not typically viewed as indexical (e.g. motion verb *gel*) can exhibit indexical properties and may be targeted by the same kind of shifting operator. This supports context-overwriting approaches to indexical shift (e.g. Anand & Nevins 2004; Deal 2020) and is not directly predicted by approaches that reduce indexical shift to syntactic agreement and pronominal feature/index manipulation (e.g. Podobryaev 2014; Messick 2023). Though the details of these accounts are different, they rely on the manipulation of complex indices/features of pronominal elements in syntax. However, considering that indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential markers) are not pronominal items, it is not directly clear how these approaches to indexical shift could account for the existence of indexical PSIs. On a shifty OP analysis on the other hand, the indexical shifting OP can target all elements that have indexical properties of the relevant type (e.g. *author*), so the shifting (and shift together) facts of the indexical PSIs are more straightforwardly explained. An alternative account has been proposed by Baker (2023), who argues that indexical shift is a result of syntactic binding/control relation between syntactic *Speaker/Adressee* heads and indexical pronominals. Though indexical PSIs do not provide any clear evidence against Baker’s approach, it faces some questions regarding how and why separate indexical elements behave differently under different embedding verbs, thus the current study concludes that the data introduced here can be best accounted for under the shifty operator accounts (e.g. Anand & Nevins 2004; Deal 2020).

5 Conclusion

In summary, this study provides novel data relevant for how perspective sensitive elements (PSIs) are interpreted in environments where indexical shift occurs. The study shows that different types of PSIs show unique properties. While most types of PSIs are not affected by indexical shift, a subtype of PSIs like motion verb *gel* ‘to come’ and evidential markers, which I call *indexical PSIs*, behave like indexicals and show shift together patterns with 1st person indexicals (e.g. they either shift together or stay non-shifted). To account for the facts introduced here, I propose that *indexical PSIs* have semantic definitions that refer to the *author* parameter of the context, and thus shift whenever indexical shift takes place.

I follow the shifty operators analysis of Deal (2020), adopted for Turkish by Oğuz et al. (2020), and suggest that context-overwriting operators (e.g. OP_{AUTH}) can merge in the left periphery of finite embedded clauses in Turkish and replace the *author* parameter of the context in which the sentence is interpreted in. For the PSIs that do not exhibit indexical-like properties, I follow existing literature which suggests that PSIs are interpreted relative to another (non-indexical) coordinate of the context such as *judge* (e.g. Lasersohn 2005).

The existence of indexical PSIs provides support for the context-overwriting approaches to indexical shift (e.g. Anand & Nevins 2004; Deal 2020) over the syntactic binding/control and feature deletion/manipulation accounts (e.g. Podobryaev 2014; Messick 2023; Ganenkov 2023; Blunier 2024), considering that these accounts rely on the binding relations between two nom-

¹⁴ A similar test cannot be applied to evidential markers, since the evidential marker ‘-mİş’ cannot attach nominalized clauses (e.g. Korotkova 2019).

inal elements and/or modifications in their feature sets. Turkish data introduced here shows for the first time that non-nominal elements such as motion verbs and evidential markers can have indexical-like properties and be targeted by indexical shift. Assuming that non-nominal elements cannot be bound, nor they have phi-features that can be manipulated, the syntactic binding/control and feature deletion/manipulation accounts are not able to straightforwardly capture the shifty behaviors of those items. In contrast, the context-overwriting approach to indexical shift (e.g. Anand & Nevins 2004; Deal 2020) do not face those weaknesses as it does not rely on direct binding/feature modification of the indexical elements. However, it is important to note that these two mechanisms are not necessarily mutually exclusive.¹⁵ It may be the case that feature deletion/manipulation mechanisms are responsible for the shifty agreement phenomenon reported in Messick (2023) and Ganenkov (2023), while the context-overwriting mechanism is responsible for indexical shift observed in Turkish and other languages.

Abbreviations

ACC = accusative, AOR = aorist, DAT = dative, DUB = dubitative, EVID = evidential, FUT = future, GEN = genitive, LOC = locative, MOD = modal, NEG = negation, NMLZ = nominalizer, OPT = optative, PL = plural, POSS = possessive, PROG = progressive, SG = singular

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To be added later.

Competing interests

The author(s) has/have no competing interests to declare.

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¹⁵ We thank an anonymous reviewer for pointing this out.

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